UNIVERSITÀ
DEGLI STUDI
DI PADOVA

MASTER'S THESIS · COMPUTATIONAL FINANCE

Academic Year 2025 / 2026

# Pricing and Calibration of Bitcoin Inverse Options via rBergomi

*A hybrid-scheme calibration study extending rough volatility evidence from BTC prices to the inverse-option surface on Deribit.*

CANDIDATE
Riccardo Caruso

SUPERVISOR
Prof. Giorgia Callegaro

DATE
22 April 2026

# Agenda

# 01 Motivation

- **STARTING POINT**

  Crypto options differ structurally from equities: standard stochastic volatility models (e.g., Heston) are only approximate—especially OTM and in turbulent regimes; Bitcoin volatility is rough ($H \in [0.05, 0.12]$, Takaishi), and inverse contracts on Deribit (BTC-denominated payoffs) require adapting classical frameworks (Black–Scholes / Heston).

- **RESEARCH QUESTION**

  *Which computational scheme for rBergomi best balances accuracy and speed in calibrating Bitcoin inverse options?*

- **CONTRIBUTION**

  - Extend Takaishi's rough-volatility evidence from BTC prices to the BTC IV surface.
  - Apply rBergomi under the inverse-option convention used on Deribit.
  - Compare three computational schemes and identify the most reliable calibration.

# 02 Data & Market

- **SOURCE**

  Deribit: ≈ 90% of global open interest.

- **INSTRUMENTS**

  European-style inverse options on BTC, cash-settled in BTC.

- **PANEL**

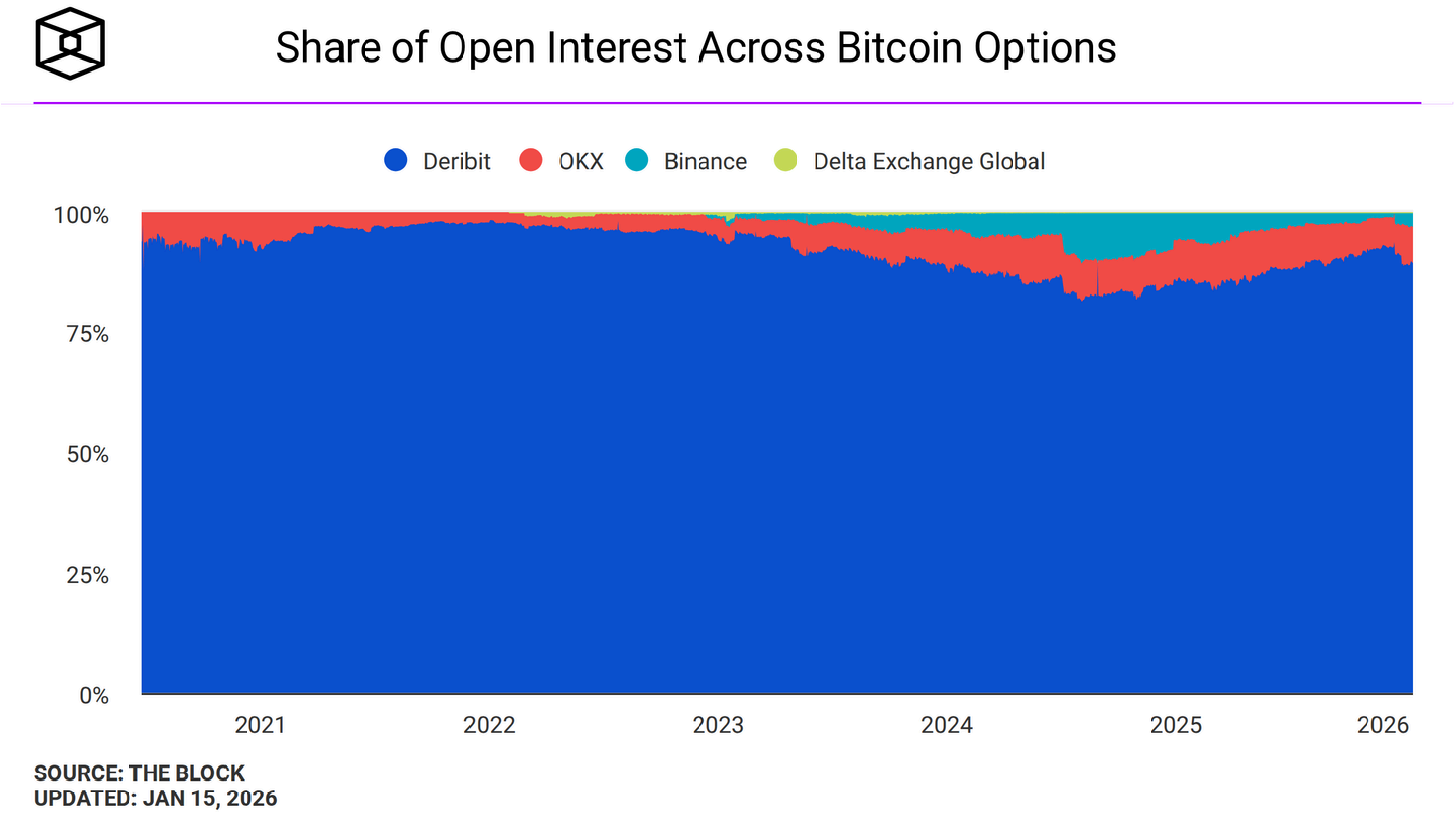
  Daily snapshots of the full option chain.

- **FILTERS**

  Remove illiquid quotes, stale mids and extreme moneyness.

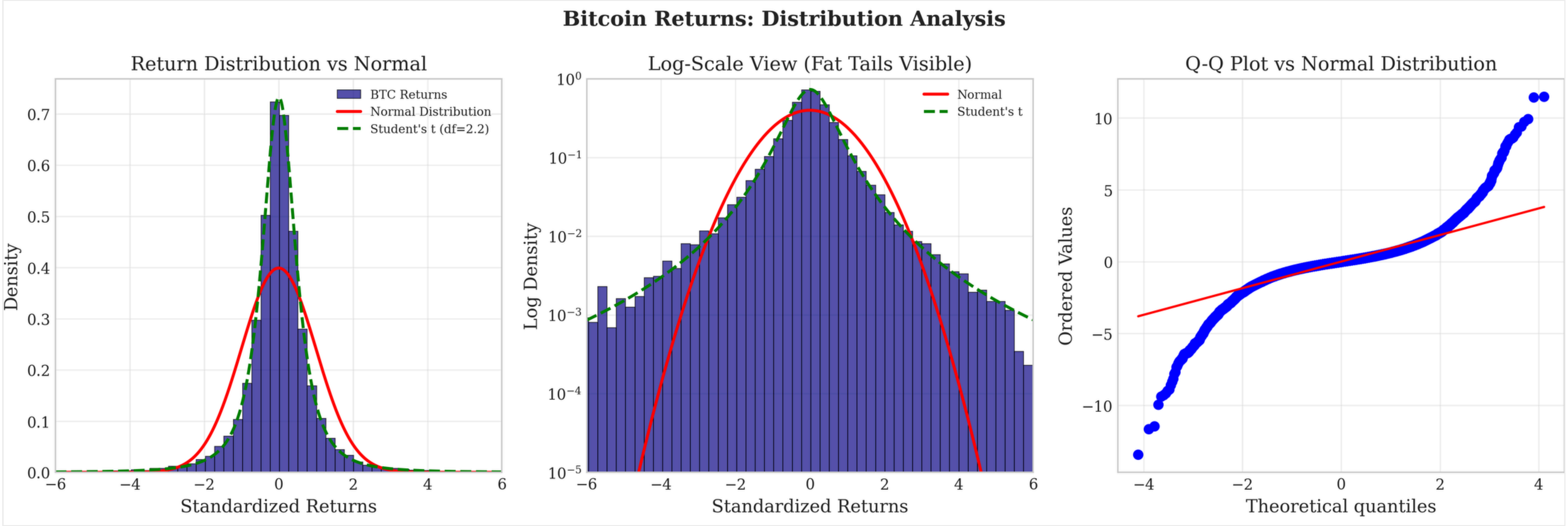
- **SPOT&RATES**

  Spot from Deribit index, zero interest-rate assumption (r=0), standard in the Deribit literature.



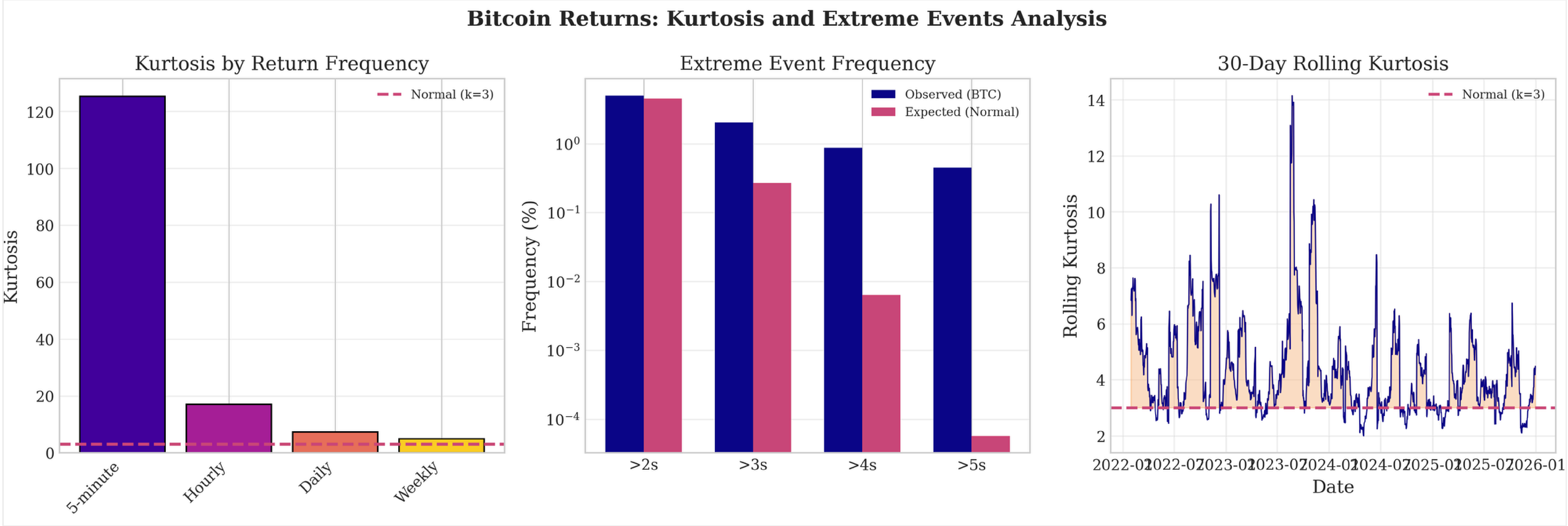
# 02 Data&Market - BTC different behavior

Fat tails, clustering, conditional heteroskedasticity.



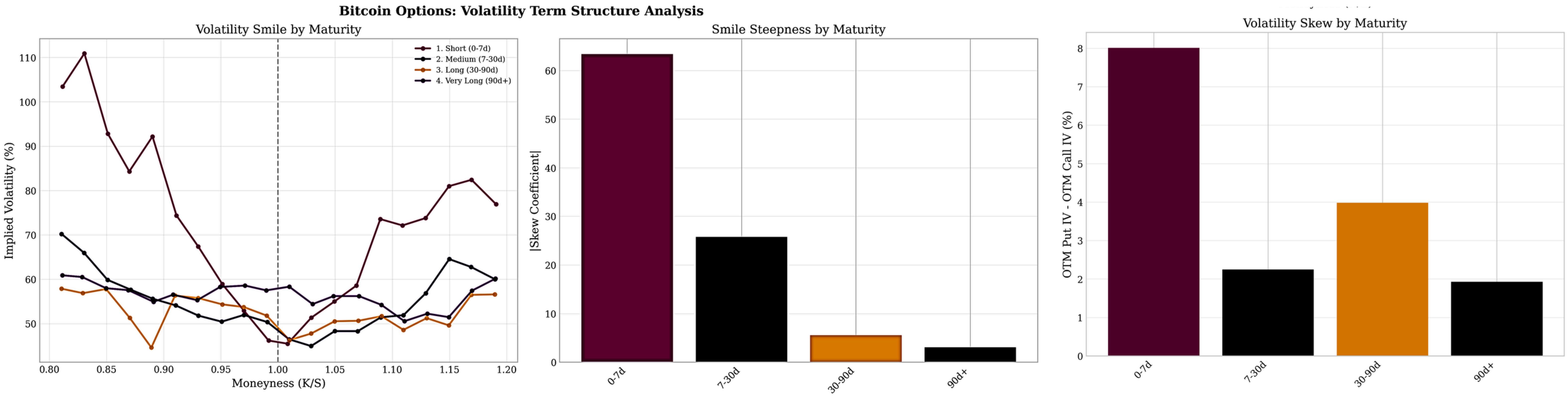
# 02 Data&Market - BTC different behavior

Spiky bursts, no mean-reversion over short horizons.



# 02 Data&Market - BTC different behavior

## Steep short-dated smile and skew flips across regimes.



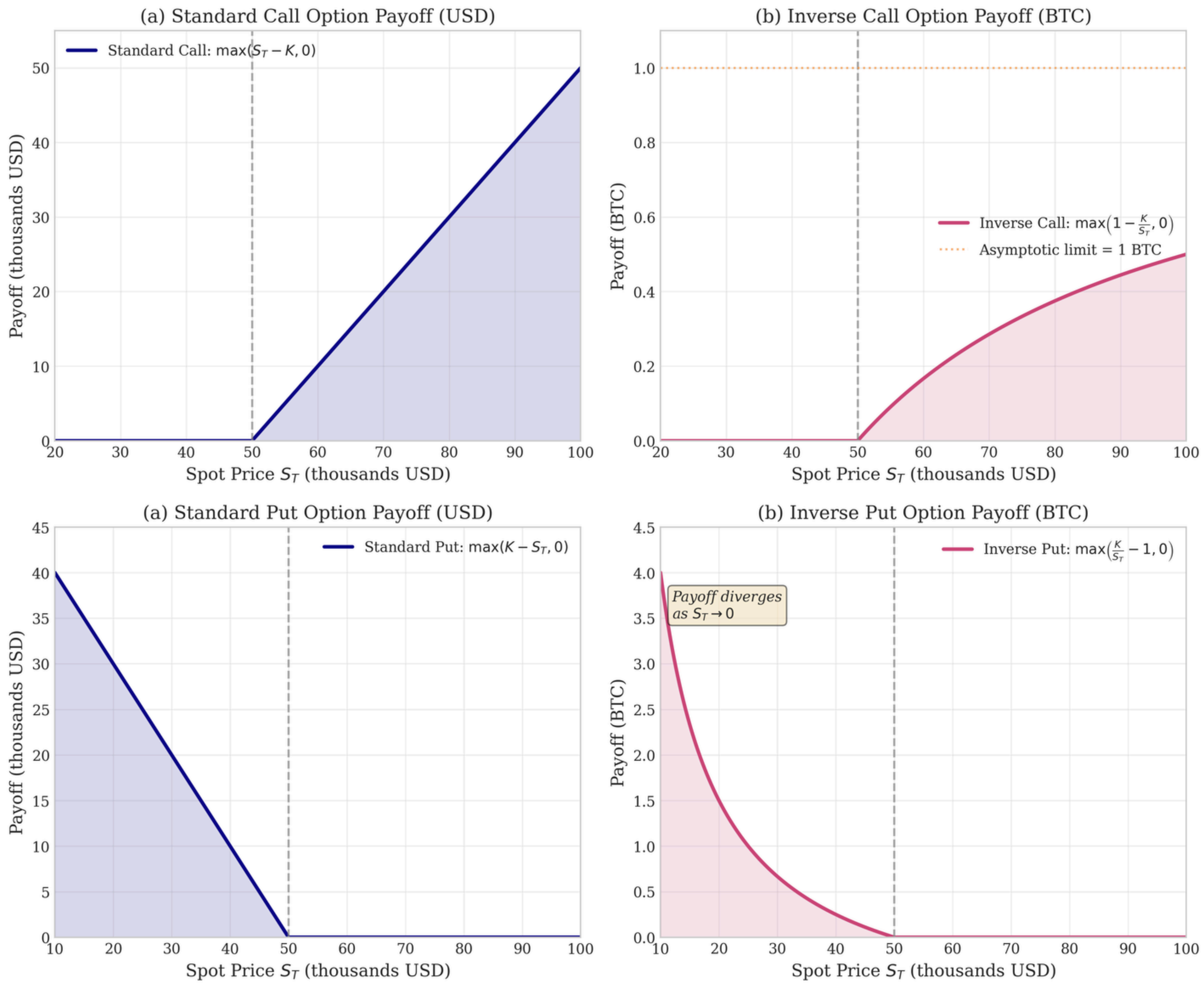

- ATM IV rises from 45.9% (0–7d) to 57.6% (90d+);
- Skew collapses from 8.0% to 1.9% across maturities — consistent with rough volatility power-law scaling;
- Moneyness ∈ [0.8, 1.2], 7–90 days, IV aggregated by VWAP.
- Short-dated BTC options exhibit the steepest smile and most asymmetric skew — patterns consistent with rough volatility dynamics and incompatible with constant-volatility or Heston-type models.

# 02 Data&Market - How Inverse Options Differ

On Deribit, BTC options are inverse: premium and payoff are denominated in BTC, not in USD.

$$C_T^{BTC} = \frac{\max(S_t - K, 0)}{S_T}$$

$$P_T^{BTC} = \frac{\max(K - S_t, 0)}{S_T}$$



# 03 The rough Bergomi model

## THREE PARAMETERS TO CALIBRATE

**H**

ROUGHNESS

*$0 < H < ½$*

**η**

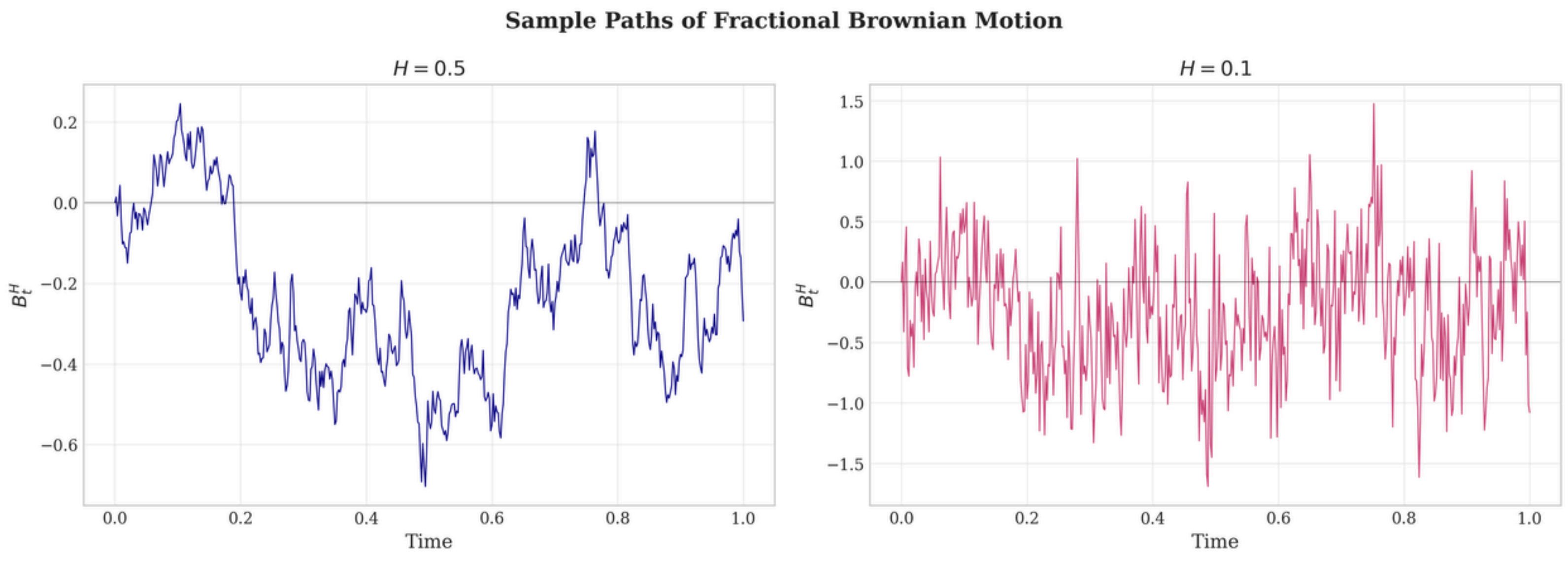
VOL-OF-VOL

*controls smile curvature*

**ρ**

SPOT / VOL

*controls smile skew*




*Fractional BM: H=0.5 (smooth) vs. H=0.1 (rough)*

Variance is driven by **Volterra process**:

$$v_t = \xi_0(t) \cdot \exp\left(\eta \tilde{W}_t - \frac{\eta^2 t^{2H}}{4H}\right)$$

$$\widetilde{W}_t = \int_0^t (t-s)^{H-1/2} dW_s^2$$

The asset price follows

$$\frac{dS_t}{S_t} = \sqrt{v_t}\, dW_t^1,\, d\langle W^1, W^2\rangle_t = \rho\, dt$$

# 03 The rough Bergomi model

> *"The Hurst parameter of Bitcoin realised variance lies in [0.05, 0.12], confirming rough volatility dynamics."*
>
> — Takaishi, *Physica A*, 2020 (paraphrased)

## THE GAP WE ADDRESS

- Takaishi tested rough volatility on the **price process**, not on IV.
- Rough variance dynamics are consistent with the steep short-end skew observed in BTC options

## HYPOTHESIS

If BTC spot dynamics are rough, then an option-level rBergomi calibration should reproduce the IV surface more tightly than Heston, especially at short maturities where roughness matters most.

## AIM

This work closes the loop: calibrate rBergomi directly to the **Deribit IV surface** and identify the most efficient computational scheme.

# 04 Calibration

| APPROACH | SCHEME | ESTIMATOR | COMPLEXITY | CHARACTERISTIC |
|---|---|---|---|---|
| **Cholesky + Euler** | Coarse-Grid Cholesky | Log-Euler | $O(n_c^3 + n \cdot n_c)$ | Exact driver; interpolation error; O(n²) memory; slowest on long grids |
| **Hybrid + Mixed** | Hybrid scheme | Mixed estimator | $O(n \cdot \kappa)$ | Lowest variance; control variate; fastest and most accurate in practice |
| **Hybrid + Euler** | Hybrid scheme<br>BLP (2017), κ=6 | Log-Euler | $O(n \cdot \kappa)$ | No interpolation; fast; high MC variance at short maturities |

Summary of the three computational approaches used in the thesis (Chapter 3).

# 04 Calibration - Variance reduction & the Mixed estimator

## STEP 1 · HYBRID DRIVER

### Simulating $v_t$ with the hybrid scheme

- Exact near-field kernel + Riemann-sum for the tail.
- The hybrid driver preserves the law of $\widetilde{W}$ at the first grid point.

## STEP 2 · CONTROL VARIATE

### Replacing the plain MC estimator

- Computing the BS price analytically under the conditional Gaussian law of $S|v_t$ .
- Eliminating the dominant source of MC noise.

## STEP 3 · MIXED ESTIMATOR

### Combine conditional and empirical

- Convex combination of the conditional analytic price and the empirical mean.
- Variance ≈ 1/10 of plain MC.

# 04 Calibration - Data handling, loss function & snapshots

## DATA CONSTRAINTS

- **Liquidity filter.** Drop quotes with zero volume or relative spread > 10%.
- **Moneyness band.** Keep 0.8 ≤ K / S ≤ 1.2 to avoid unreliable deep wings.
- **Maturity grid.** Maturities 7–90 days — weekly, monthly, quarterly.
- **Snapshots, not history.** Each calibration uses a *single* option-chain snapshot.
- **Chosen dates.** 30 snapshots: 9 baseline (3 regimes × 3 dates) + 21 event dates (7 events × 3 days).

## LOSS FUNCTION

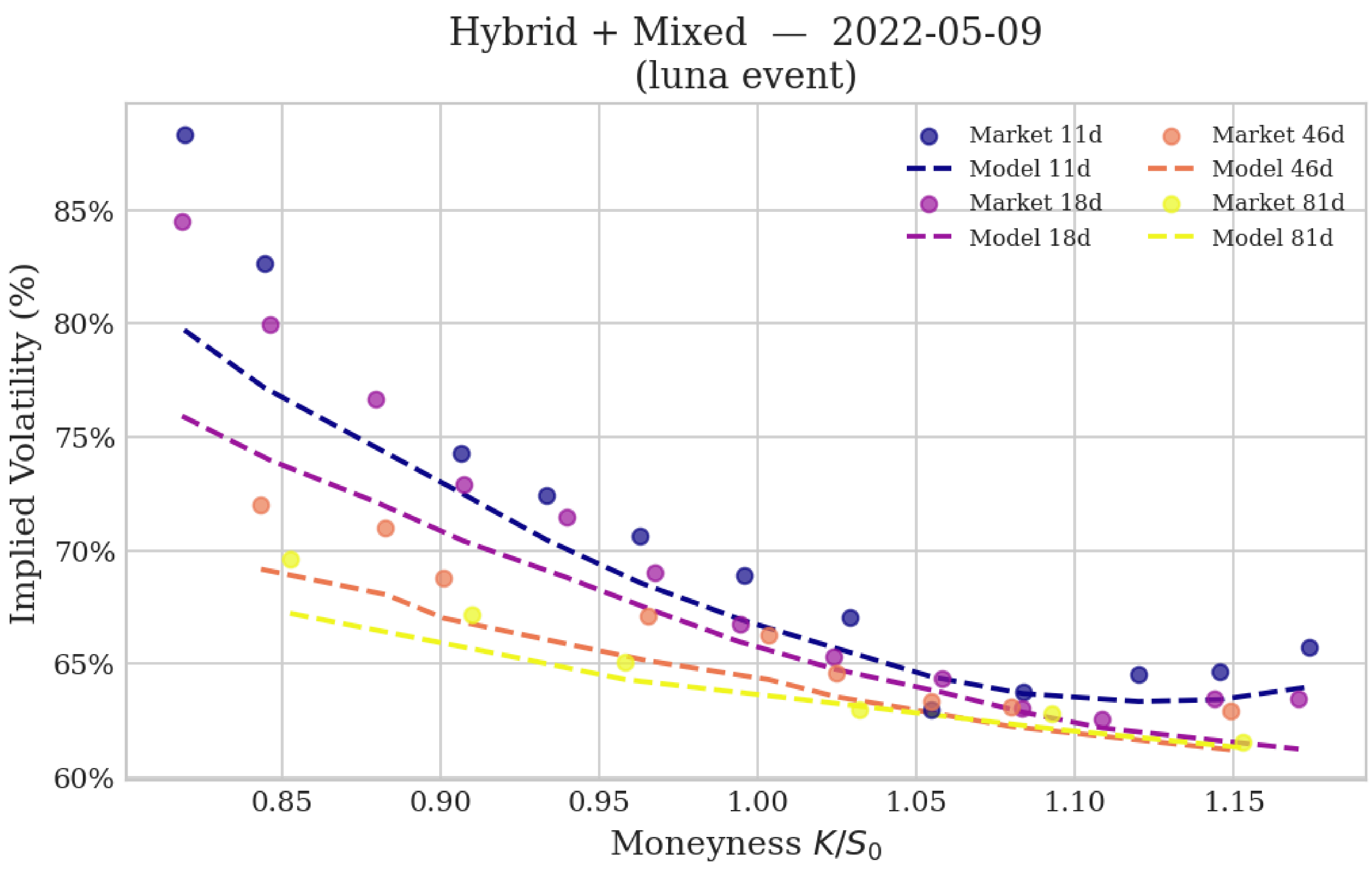
$$\mathcal{L}(\boldsymbol{\theta}) = \sqrt{\sum_{i=1}^{N} w_i \left(\sigma_i^{\mathrm{mod}}(\boldsymbol{\theta}) - \hat{\sigma}_i\right)^2} \times 100$$

- **IV-space, not price-space.** More stable across strikes than BTC price-space errors.
- **Maturity-weighted.** $w_i \propto 1/\sqrt{T_i}$ to up-weight short-dated options, where the ATM skew is most sensitive to H.
- **Optimiser.** Differential evolution for a global pass, followed by local Nelder–Mead refinement.
- **$\xi_0(t)$.** Bootstrapped from the ATM term structure; calibrated independently per snapshot.

# 05 RESULTS - HIGH IV SNAPSHOT

LUNA COLLAPSE · 09-MAG-2022

| SCHEME | RMSE IV | TIME |
|---|---|---|
| Hybrid + Euler | 32.93 pp | 42 s |
| Hybrid + Mixed | 27.23 pp | 17 s |
| Cholesky + Euler | 73.25 pp | 7.1 min |

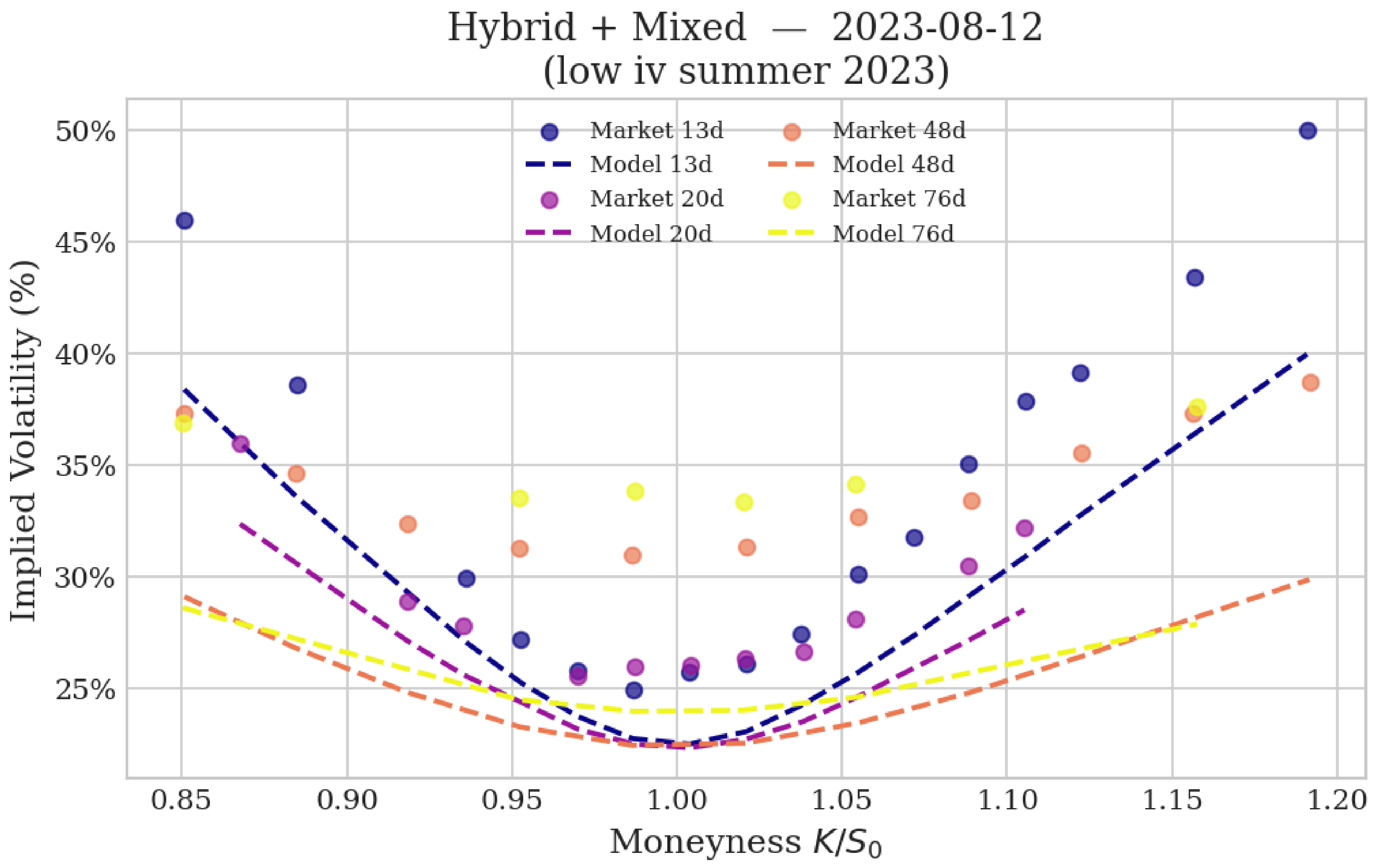



→ At the extreme event snapshot, Mixed beats both plain Hybrid (noise dominates) and Cholesky (at the same sample size, the variance reduction wins).

# 05 RESULTS - LOW IV SNAPSHOT

CALM MARKET · 12-AUG-2023

| SCHEME | RMSE IV | TIME |
|---|---|---|
| Hybrid + Euler | 2.97 pp | 37 s |
| Hybrid + Mixed | 3.94 pp | 9 s |
| Cholesky + Euler | 3.10 pp | 1.9 min |

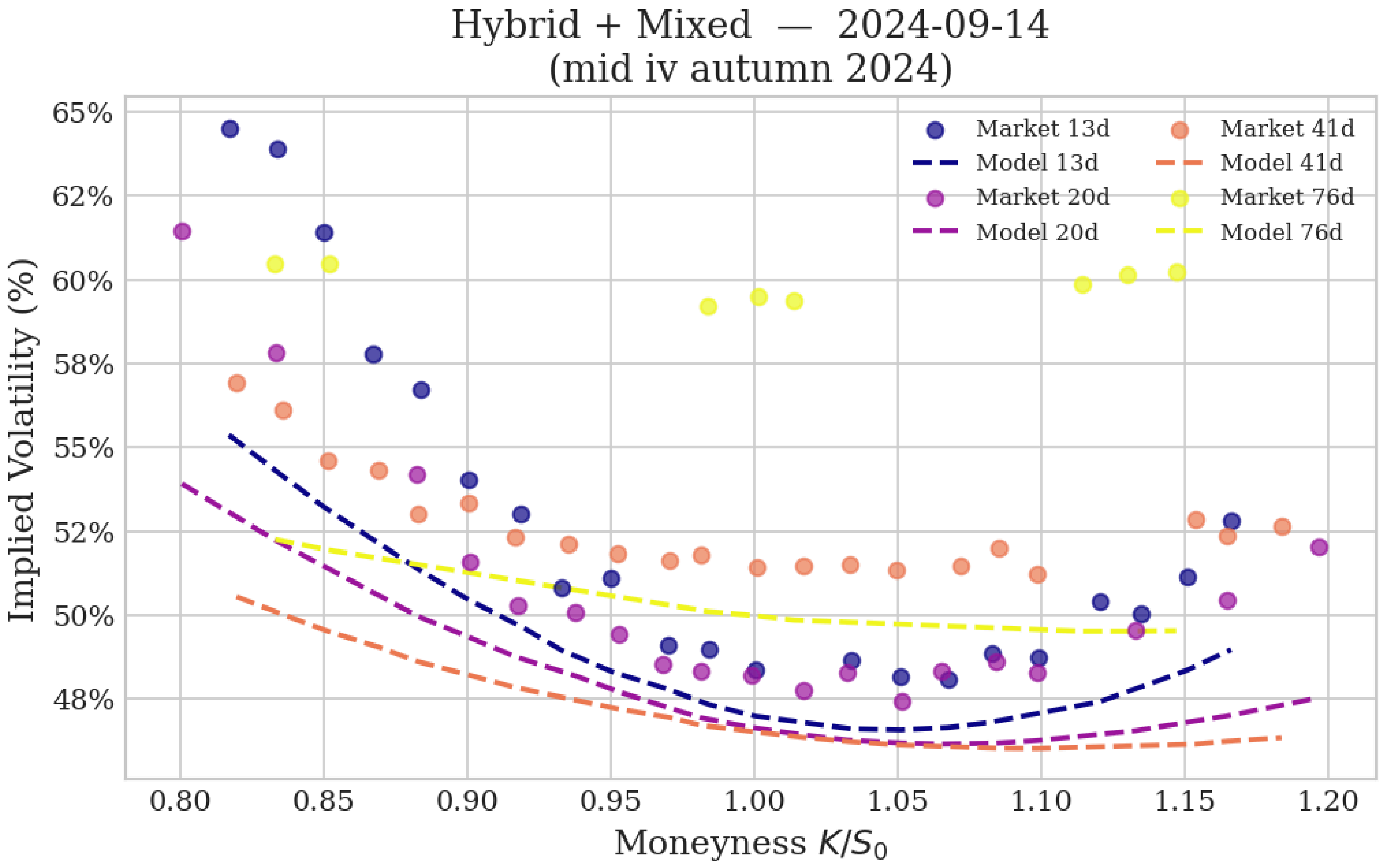



→ In calm markets, all three schemes achieve comparable accuracy — the Mixed estimator's advantage is purely in speed.

# 05 RESULTS - MEDIUM IV SNAPSHOT

AUTUMN · 14-SEP-2024

| SCHEME | RMSE IV | TIME |
|---|---|---|
| Hybrid + Euler | 78.00 pp | 23 s |
| Hybrid + Mixed | 11.45 pp | 20 s |
| Cholesky + Euler | 22.50 pp | 5.7 min |




→ In medium-stress conditions, Hybrid + Mixed outperforms Cholesky by a factor of two in accuracy (11.45 vs 22.50 pp) at a fraction of the elapsed time.

# 05 REGIMES - LOW IV

| DATE | SCHEME | H | η | ρ | RMSE IV | TIME |
|---|---|---|---|---|---|---|
| 12/08/2023 | Hybrid + Euler | 0,1312 | 2,3080 | -0,1940 | 2,97 pp | 37 s |
| | Hybrid + Mixed | 0,0100 | 2,7010 | 0,0000 | 3,94 pp | 9 s |
| | Cholesky + Euler | 0,0835 | 2,2280 | -0,3060 | 3,10 pp | 1,9 min |
| 30/09/2023 | Hybrid + Euler | 0,1622 | 2,2990 | -0,5260 | 4,20 pp | 24 s |
| | Hybrid + Mixed | 0,0266 | 2,6030 | -0,0110 | 5,5 pp | 15 s |
| | Cholesky + Euler | 0,0785 | 2,0750 | -0,6980 | 3,82 pp | 1,5 min |
| 10/02/2024 | Hybrid + Euler | 0,0100 | 2,8810 | -0,0660 | 34,95 pp | 21 s |
| | Hybrid + Mixed | 0,0100 | 2,8810 | -0,0660 | 7,34 pp | 12 s |
| | Cholesky + Euler | 0,0418 | 2,5780 | -0,3840 | 8,44 pp | 1,6 min |

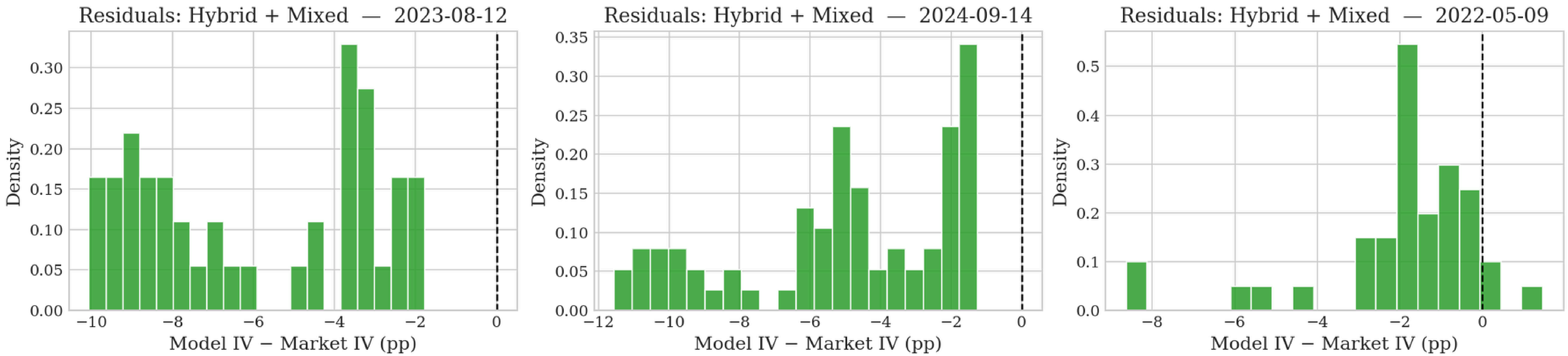
# 05 REGIMES - MEDIUM IV

| DATE | SCHEME | H | η | ρ | RMSE IV | TIME |
|---|---|---|---|---|---|---|
| 17/12/2022 | Hybrid + Euler | 0,3171 | 2,6920 | -0,9900 | 15,54 pp | 14 s |
| | Hybrid + Mixed | 0,0100 | 2,0860 | -0,4030 | 58,10 pp | 26 s |
| | Cholesky + Euler | 0,2318 | 3,8050 | -0,7870 | 20,29 pp | 2,1 min |
| 02/07/2022 | Hybrid + Euler | 0,4308 | 3,5650 | -0,8810 | 29,55 pp | 2,0 min |
| | Hybrid + Mixed | 0,0100 | 0,9270 | -0,7910 | 38,10 pp | 14 s |
| | Cholesky + Euler | 0,0239 | 3,6200 | -0,9380 | 90,17 pp | 3,0 min |
| 14/09/2024 | Hybrid + Euler | 0,0100 | 3,2310 | 0,0000 | 78,00 pp | 23 s |
| | Hybrid + Mixed | 0,0100 | 3,2310 | -0,2230 | 11,45 pp | 20 s |
| | Cholesky + Euler | 0,0436 | 2,9990 | -0,6560 | 22,50 pp | 5,7 min |

# 05 REGIMES - HIGH IV

| DATE | SCHEME | H | η | ρ | RMSE IV | TIME |
|---|---|---|---|---|---|---|
| 18/06/2022 | Hybrid + Euler | 0,0594 | 3,6330 | -0,6650 | 65,97 pp | 16 s |
| | Hybrid + Mixed | 0,0139 | 1,4130 | -0,9900 | 88,61 pp | 2,2 min |
| | Cholesky + Euler | 0,0245 | 3,5870 | 0,0000 | 102,62 pp | 1,9 min |
| 16/11/2022 | Hybrid + Euler | 0,1520 | 3,8470 | -0,8080 | 53,81 pp | 1,1 min |
| | Hybrid + Mixed | 0,0100 | 1,5650 | -0,7800 | 27,92 pp | 12 s |
| | Cholesky + Euler | 0,2098 | 2,9460 | -0,8380 | 38,53 pp | 5,6 min |
| 03/02/2025 | Hybrid + Euler | 0,0100 | 3,7080 | 0,0000 | 37,46 pp | 22 s |
| | Hybrid + Mixed | 0,0100 | 1,6760 | 0,0000 | 20,20 pp | 17 s |
| | Cholesky + Euler | 0,0644 | 3,3000 | -0,6740 | 20,34 pp | 3,9 min |

# 05 RESIDUALS




- **Low volatility —** smallest overall fit error (RMSE ≈ 4 pp); moderate spread (σ ≈ 2.8 pp); mild systematic underestimation.
- **Medium volatility —** moderate residuals; slight wing curvature mismatch at short maturities.
- **High volatility —** wider residuals at short-dated wings; model underestimates extreme skews due to insufficient curvature and H pinned at lower bound.

→ Across regimes, **rBergomi performs best in calm markets**, where smooth variance dynamics closely match the observed surface. Fit quality degrades progressively with volatility stress.

# 06 CONCLUSIONS

- Rough volatility extends to options: Takaishi's price-level evidence carries over to the BTC option surface (mean $H \approx 0.063$ across regimes, often near the lower bound).
- Hybrid + Mixed is the recommended scheme: half of the RMSE of Cholesky+Euler (22.83 pp vs 41.76 pp) at a fraction of the computational cost.
- Best fit in calm markets: rBergomi delivers the tightest residuals in low-vol regimes, with mild underfitting in short-dated wings under stress.

REPRODUCIBILITY

`github.com/RVJIM/rBegromi_Bitcoin_Option`

All calibration code, data pipelines and figures are open-source and reproducible.

# Thank you for your attention.

*Questions are welcome.*